S2.1 The ionic model

Formation of ions

Learning outcomes

By the end of this section you should be able to:

- Identify cations and anions from their neutral atom.
- Deduce the charge and symbol for ions.
- Deduce the electron configuration of a species based on its ionic symbol.

In section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) you learned that elements are the primary constituents of matter, which cannot be chemically broken down into simpler substances. Growing up, you may have learned that iron is important for your diet and you can likely name some foods that are rich in iron, for example, meat, beans and dark leafy greens. You may also have learned that iron is a metal used to create car bodies, bridges and skyscrapers (**Figure 1**). When eating iron rich foods, do you see or eat pieces of metal?

Credit: Anjelika Gretskaia, Getty Images (https://www.gettyimages.co.uk/detail/photo/fresh-spinach-leaves-on-white-background-royalty-free-image/1369248724)

Credit: tunart, Getty Images (https://www.gettyimages.co.uk/detail/photo/the-iron-bridges-of-chicago-downtown-royalty-free-image/1447903626)

Figure 1. Examples of foods and structures containing iron.

In section S1.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/) you learned that all atoms of the same element have the same number of protons. You also learned that isotopes of the same element will have the same chemical properties, even though the number of neutrons differ. What subatomic particle could be responsible for the difference between the iron in foods we eat versus the iron we use in building materials? In section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) you learned about the electron configuration of atoms. Can there be a change to the number of electrons that might result in different properties for the same element?

Everything is made up of atoms, yet these atoms rarely exist as individual atoms. Atoms usually interact with other atoms to form chemical bonds. Chemical bonds create a more stable electron configuration for the atoms involved.

In previous sections, you have learned about elements and the nuclear and electronic components of the atom. Elements are the primary constituents of matter. Elements can be classified as metals or non-metals. **Figure 2** shows a periodic table classifying metals and non-metals. Note that there is also a region in between the two that are classified as metalloids. These are elements that show intermediate properties between metals and non-metals.

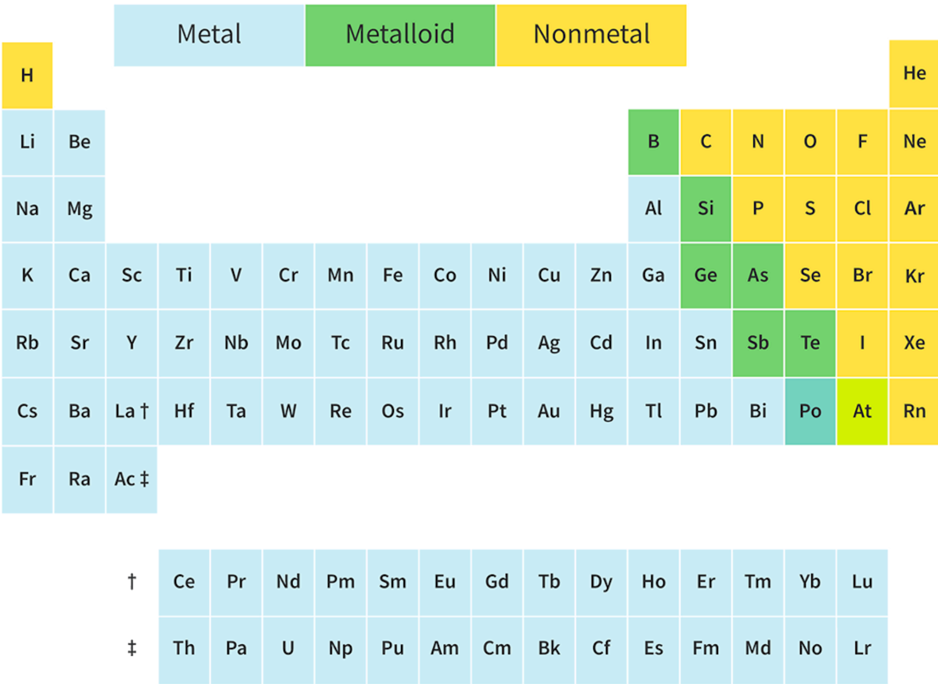


Figure 2. Classification of metals, non-metals and metalloids on the periodic table.

More information for figure 2

This diagram displays the periodic table of elements, categorizing them into metals, nonmetals, and metalloids. The metals are shaded in light blue, occupying the majority of the table, particularly on the left and center. Nonmetals are highlighted in yellow on the top right, including elements like hydrogen (H), carbon (C), nitrogen (N), and oxygen (O). Metalloids, bordered in green, sit between metals and nonmetals, featuring elements such as boron (B), silicon (Si), and arsenic (As). The diagram provides a visual representation of the classification of elements based on their properties.

[Generated by AI]

When looking at electron configurations in section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) what configurations were the most stable? Did you notice that these stable electron configurations occur when valence electrons occupy an energy level that is completely filled? What about atoms that have valence electrons occupying an energy level that is not completely filled? How might these atoms become more stable?

Metallic elements have low numbers of electrons in the valence shells. How do you think a metallic element might become more stable? Non-metals have a greater number of electrons in the valence shell. How do you think a non-metal might become more stable? A metal can lose electrons or a non-metal can gain electrons in order to become more stable forming an ion.

Exercises 1 and 2

Click a question to answer

1. ________________ has the electron configuration $1s^2 2s^2 2p^6$.

________________ has the unstable electron configuration $1s^2 2s^2 2p^6 3s^1$ or [Ne] $3s^1$.

2. True or false?

Neon has a stable electron configuration.

Check answers

Ions

An ion is an electrically charged species that is formed after an atom gains or loses an electron.

- When metal atoms lose electrons, they form positive ions called cations. A cation has fewer electrons than protons.
- When non-metal atoms gain electrons, they form negative ions called anions. An anion has more electrons than protons.

The number of protons and electrons in an atom are equal, and the atom is electrically neutral. The number of protons and electrons in an ion is not the same and the ion carries an overall charge.

Figure 3a shows a comparison of the electron structure for a neutral sodium atom and a sodium cation. **Figure 3b** shows a comparison of the electron structure for a neutral chlorine atom and a chloride anion (not to scale).

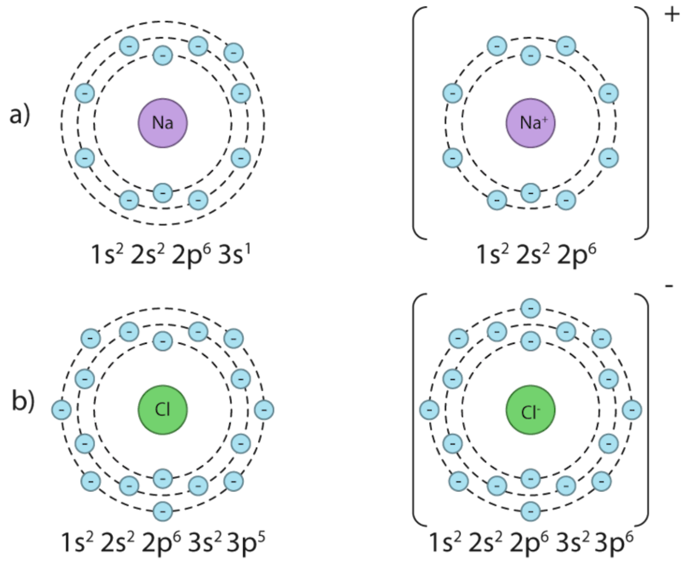


Figure 3. Comparison of the electron configuration of neutral atoms to ions: a) cations and b) anions.

More information for figure 3

The image shows two sets of electron configuration diagrams. In diagram a), on the left, a neutral sodium atom is depicted. It's represented by a central circle labeled 'Na' surrounded by circular shells showing electron distribution: $1s^2$, $2s^2$, $2p^6$, $3s^1$. On the right of this is a sodium cation (Na^+), which shows a central circle labeled 'Na^+', with a changed electron distribution excluding the outer electron: $1s^2$, $2s^2$, $2p^6$.

In diagram b), on the left, a neutral chlorine atom is represented by a central circle labeled 'Cl', with its electrons distributed in the shells as: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^5$. On the right is a chloride anion (Cl^-) where the central circle is labeled 'Cl^-', with an added electron, completing the outer shell: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$.

[Generated by AI]

Interactive 1 shows a visual representation of the formation of anions and cations.

Interactive 1. Visual representation of the formation of cations.

More information for interactive 1

An interactive animation model depicts the creation of a sodium cation. It illustrates the atomic structure of a sodium atom, displaying 11 electrons surrounding the nucleus and 11 neutrons in the nucleus. A seek bar with a play/pause button is at the bottom.
Selecting the play option shows an electron leaving the sodium atom, transforming it into a sodium ion (Na^+). A toggle button for repeat and skip forward is at the right of the slider.
A panel below the title-cation creation on the left provides view options, including the entire model, electrons, and protons. Selecting electrons or protons reveals their arrangement within the structure. On the right side, the contrast or dark mode and font settings are available. At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction reads as follows:
Cations are positively charged particles, which can be either atoms or molecules. They have one or more electrons fewer than their neutral form. The model depicts the formation of sodium cations, according to the reaction: $Na \rightarrow Na^+ + e^-$
Ionic compounds are composed of both cations (positively charged ions) and anions. These ions are held together by strong electrostatic forces of attraction.
The model illustrates the transformation of a sodium atom into a sodium cation and depicts the electron arrangement before and after ion formation.

Interactive 2. Visual representation of the formation of anions.

More information for interactive 2

An interactive animation model depicts the creation of a chloride anion. It illustrates the atomic structure of a chlorine atom, displaying 17 electrons surrounding the nucleus and 17 neutrons in the nucleus. A seek bar with a play/pause button is at the bottom. Selecting the play option shows an electron approaching from the top left, entering the chlorine atom, transforming it into a chlorine ion (Cl^-). A toggle button for repeat and skip forward is at the right of the slider.
A panel below the title- anion creation on the left provides view options, including the entire model, electrons, and protons. Selecting electrons or protons reveals their arrangement within the structure. On the right side, the contrast or dark mode and font settings are available.
At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction reads as follows:
Anions are negatively charged particles, which can be either atoms or molecules. They have one or more extra electrons than their neutral form. The model depicts the formation of chlorine anions, according to the reaction: $Cl + e^- \rightarrow cl^-$
Ionic compounds are composed of both cations (positively charged ions) and anions. These ions are held together by strong electrostatic forces of attraction.
The model illustrates the transformation of a chlorine atom into a chloride anion and depicts the electron arrangement before and after ion formation.

Making connections

When an ion is formed, its radius changes from its atom. The ionic radius of an anion is always larger than its neutral atomic radius and the ionic radius of a cation is always smaller than its neutral atomic radius. This is covered in section S3.1.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/).

The octet rule refers to the tendency of an atom to have a valence shell with a total of 8 electrons. You will find more detail about this rule in section S2.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/). This rule is helpful in deducing how many electrons an atom will lose or gain in order to become electronically stable. **Table 1** shows the deduction of ionic charge for Period 3 elements. Note that this pattern is applicable to the entire group.

Table 1. Deducing the ionic charge for Period 3 elem

Group number	**1**	**2**	**13**	**14**	**15**	**16**
Period 3 elements	**Na**	**Mg**	**Al**	**Si**	**P**	**S**
Electron configuration	[Ne] $3s^1$	[Ne] $3s^2$	[Ne] $3s^2 3p^1$	[Ne]$3s^2 3p^2$	[Ne] $3s^2 3p^3$	[Ne] $3s^2 3p^4$
Number of electrons in valence shell	1	2	3	4	5	6
Electron gain/ loss	Lose 1	Lose 2	Lose 3	Unlikely to form ions	Gain 3	Gain 2
Ion formed	Na^+	Mg^{2+}	Al^{3+}	**N/A**	P^{3-}	S^{2-}
Electron configuration of ion	[He] $2s^2 2p^6$	[He] $2s^2 2p^6$	[He] $2s^2 2p^6$		[Ne] $3s^2 3p^6$	[Ne] $3s^2 3p^6$

Since sodium has one valence electron, it is energetically more favourable to lose one electron than to gain seven to achieve a full valence shell. After losing one electron, it will have more protons in the nucleus than electrons surrounding it.

This will give it an overall charge of 1+. Notationally, the symbol for an ion is represented by the element symbol first, followed by a superscript to show the charge. In the superscript, the number is always written first, followed by the charge. This means that when the neutral element Na loses an electron, it would be represented by Na^{1+}, or more commonly Na^{+}.

The box below has some important things to consider for proper notation of ions.

Study skills

Notation for ions

1. (a) Ions are expressed using elemental symbols stated in the guide and section 7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) of the DP chemistry data booklet. The charge of an ion can be deduced by considering the ratio of protons to electrons.
   (b) When ions form, it is electrons that are either lost or gained. The number of protons is **always** the same for a particular atom.
2. The number is **always** written first, followed by charge for the symbolic form of an ion, e.g. Mg^{2+}. When discussing the term oxidation state the charge is written first, e.g. The Mg^{2+} ion has a +2 oxidation state.
3. For an ion with a charge of +1 or —1, the 1 does not need to be shown in the element symbol. For example, lithium forms an ion with a charge of 1+. The symbol for this ion would be Li^{+}.

As sulfur has six valence electrons, it is energetically more favourable to gain two electrons than to lose six. After gaining two electrons, it will have more electrons surrounding it than protons in the nucleus, giving it an overall charge of 2–. This can be represented as S^{2-}. This means that you can predict the magnitude and sign of the charge of an ion that is formed from an atom of an element in a specific group.

Nature of Science

Aspect: Patterns and trends

The placement of atoms within groups in the periodic table allows students and scientists to easily deduce how many valence electrons each element has, and therefore, the ion that they are likely to form (or even if they are likely to form an ion). Atoms that form more than one ion with different charges are also located within the same region of the periodic table.

You might have noticed that in **Table 1** silicon and argon were not assigned a charge. This is because silicon has four valence electrons, giving it a low tendency to form ions. Argon already has a full valence shell with a stable electron configuration and also has a low tendency to form ions.

Transition elements

Transition elements are those located in groups 3–12 on the periodic table. In section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) you learned how to deduce electron configuration of elements and ions up to $Z = 36$. What sublevel are valence electrons occupying for $Z = 21$ to $Z = 30$? Recall how the 3d sublevel is very close in energy to the 4s and 4p sublevels (illustrated in **Figure 4**). This allows electrons from both the 4s and 3d to act as valence electrons for elements $Z = 21$ to $Z = 30$.

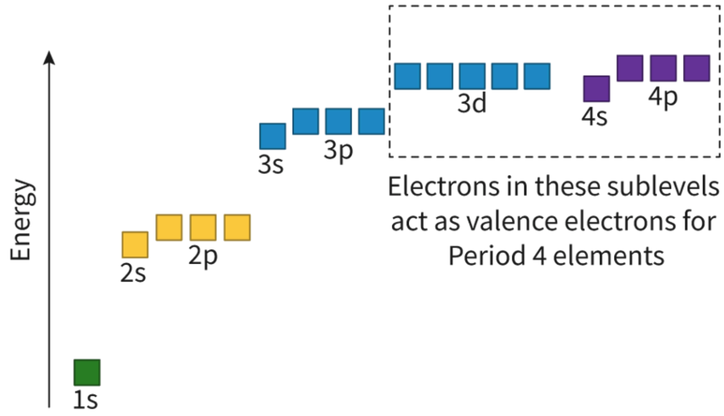


Figure 4. Relative energies of sublevels of the atom.

More information for figure 4

The image is a diagram illustrating the relative energies of atomic sublevels ranging from 1s to 4p. The diagram consists of horizontal blocks at different energy levels on a vertical axis labeled 'Energy.' Starting from the bottom, there is a single green block labeled '1s,' followed by two yellow blocks labeled '2s' and '2p.' Above these are three blue blocks labeled '3s,' '3p,' and '3d.' At the top, there are two purple blocks labeled '4s' and '4p.' The 3d, 4s, and 4p blocks are enclosed in a dashed rectangle, with a note indicating that electrons in these sublevels act as valence electrons for Period 4 elements. The blocks and their alignment are used to visually represent the close energy relationship shared among the sublevels 4s, 3d, and 4p.

[Generated by AI]

When transition elements in Period 4 lose electrons to form a cation, electrons are first lost from the 4s sublevel followed by the 3d sublevel. These two different sublevels where electrons are lost result in some variability in the charge of the ion formed for transition elements. Let's consider two possible titanium ions below:

- Ti^{2+} two electrons lost from the 4s sublevel.

- Ti^{4+} two electrons lost from the 4s sublevel and two electrons lost from the 3d sublevel.

The interactive below shows the comparison of the electron configurations between neutral titanium and the two ions discussed above.

Select, using the buttons, the different types of titanium. Look at each electron configuration and orbital diagram in **Interactive 2**.

Interactive 2. Electron configurations and orbital diagrams for Ti, Ti^{2+} and Ti^{4+}.

More information for interactive 2

An interactive tool shows orbital diagrams for Ti, Ti2+, and Ti4+, along with their electronic configurations. Three buttons at the bottom, labeled Ti, Ti2+, and Ti4+, each have a corresponding checkbox. Selecting a titanium species displays the appropriate orbital diagram and electron configuration.
A vertical arrow on the left represents energy. The energy level increases in the following order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, and 4p. In the orbital diagram, the s orbital has a single box, the p orbital has three boxes, and the d orbital has five boxes. Electrons are represented by arrows inside these boxes. Each box represents an orbital. The 3d, 4s, and 4p orbitals are represented as valence electrons.
Titanium (Ti) has 22 electrons, with an electron configuration of 1s2 2s2 2p6 3s2 3p6 4s2 3d2. The 1s, 2s, 3s, and 4s orbitals have two electrons in opposite spins, 2p and 3p orbitals have six electrons, with each box having two electrons in opposite directions, and the 3d orbital has two electrons in the first two boxes. The 4p orbital is empty.
By selecting the Ti2+, its orbital diagram and electronic configuration are displayed. Titanium (Ti 2+) has 20 electrons, with an electron configuration of 1s2 2s2 2p6 3s2 3p6 3d2. Two electrons are removed from the 4s orbital of the neutral titanium atom.

By selecting the Ti 4+, its orbital diagram and electronic configuration are displayed. Titanium (Ti 4+) has 18 electrons, with an electron configuration of 1s2 2s2 2p6 3s2 3p6. Two electrons are removed from the 4s orbital, and two electrons are removed from the 3d orbital of the neutral titanium atom.
This interactive tool helps users understand the electron arrangement in neutral titanium (Ti) and its ions (Ti^{2+} and Ti^{4+}).

Table 2 shows the electron configurations for some of the common ions formed from transition elements in Period 4.

Table 2. Electron configuration and common ionic charges for Period 4 transition eleme

Element	Electron configuration of neutral element	Common ions	Common i
Cr	$[Ar]4s^1 3d^5$	Cr^{2+} Cr^{3+}	$[Ar]3d^4$ $[Ar]3d^3$
Fe	$[Ar]4s^2 3d^6$	Fe^{2+} Fe^{3+}	$[Ar]3d^6$ $[Ar]3d^5$
Cu	$[Ar]4s^1 3d^{10}$	Cu^{+} Cu^{2+}	$[Ar]3d^{1C}$ $[Ar]3d^9$

Use the following activity to test your understanding of electron configuration of ions.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:**
  - Thinking skills
    - Critically evaluating information to reach a conclusion
    - Metacognitive awareness of how understanding is applied
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity to start, followed by work in pairs

Download and complete the following worksheet.

Worksheet
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

Once you have completed the worksheet, compare your chart with a peer. Discuss with your peer which rows you found straightforward to complete and which rows you found difficult to complete. Why do you think some rows were easier than others?

S2.1 The ionic model

Ionic bonding

Learning outcomes

By the end of this section you should be able to:

- Define ionic bonding.
- Deduce the formula and name of an ionic compound from its component ions.
- Interconvert names and formulas of binary ionic compounds.

It is important to know what the charge of ions are within a compound. Some species, such as transition elements, can form ions with different charges. The different charged ions can have completely different properties even though they are the same element!

Fehling's reagent is an example of a solution containing copper ions that is used to detect glucose in urine as a tool to screen for diabetes. The original solution is a blue, homogeneous mixture containing Cu^{2+} ions. When the solution comes in contact with glucose a reaction occurs converting the Cu^{2+} ions to Cu^{+} ions. These Cu^{+} ions form Cu_2O and create a heterogenous mixture indicating the presence of glucose in solution. This process is shown in **Figure 1**.

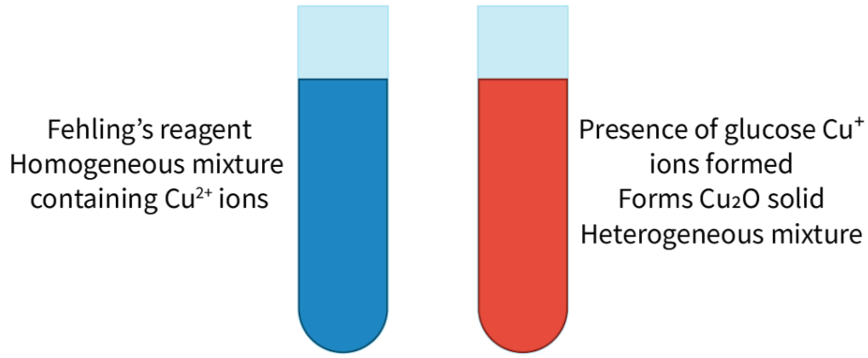


Figure 1. Fehling's reagent is used to detect the presence of glucose.

Another example of ions with varying charges is chromium. Chromium can exist as different ions such as Cr^{3+} and Cr^{6+}. In small quantities, Cr^{3+} is an essential nutrient found in foods that is required for sugar and fat metabolism in mammals. Cr^{6+}, however, is toxic and carcinogenic. Industrial pollution has caused contamination of some groundwater supplies with Cr^{6+} globally. One well known example of Cr^{6+} contamination was the Hinkley groundwater contamination made famous by the film *Erin Brockovich*. You can read more about this example here (https://www.scientificamerican.com/article/chromium-water-cancer/).

What happens when a copper or chromium cation encounters an anion? How can we look at a chemical formula, such as Cu_2O or CrO_3 and know whether the chromium ion within it is the one that acts as an essential nutrient or the toxic one?

Recall from section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) that when metal atoms lose electrons they form positive ions called cations and when non-metal atoms gain electrons they form negative ions called anions. What do you think happens if a metal is in the presence of a non-metal? The metal element can lose electrons and the non-metal element can gain the lost electrons. This process of electrons being transferred from one atom and gained in another happens simultaneously and forms ions of opposite charge. What do you think happens when ions of opposite charge are in close proximity to one another? Oppositely charged ions experience an electrostatic attraction. This is what occurs in the formation of an ionic bond and is illustrated in **Interactive 1**.

Press the ‘next’ button to watch the formation of an ionic bond.

Interactive 1. Formation of the ionic bond.

More information for interactive 1

An interactive slideshow with five slides demonstrates the ionic bonding between Sodium and Chlorine. Next and Previous buttons are provided on the bottom, allowing users to navigate between the slides.
Slide 1: An outline of the periodic table with highlighted Sodium (metal) on the left side and Chlorine (non-metal) on the right side. Sodium is in Group 1 and Period 3. Chlorine is in Group 17 and Period 3.

Slide 2: Sodium is depicted as a large circle and chlorine as a small circle. The electronic configuration of sodium is [Ne] 3s1. The electronic configuration of chlorine is [Ne] 3s2 3p5. Sodium loses one electron to have a full valence shell, and Chlorine gains one electron to have a full valence shell.
Slide 3: An animation explains how sodium transfers its valence electron to chlorine. One electron is transferred from sodium to chlorine. Ions are formed after the transfer of electrons occurs. Sodium becomes a cation (Na^+) and is represented as a small circle, while chlorine becomes an anion (Cl^-) and is represented as a large circle.
Slide 4: An animation illustrates how these oppositely charged ions experience an electrostatic attraction to one another. The Na^+ ion is attracted to the Cl^- ion. The electrostatic attraction between the oppositely charged ions is called the ionic bond.
Slide 5: The Na^+ ion interacts with the Cl^- ion.
A play/pause button is available on the bottom left to control the animations. An option for full screen is available at the bottom right.
This interactivity helps users understand the formation of NaCl through ionic bonding and the electrostatic attraction between sodium and chloride ions.

The ionic bond is formed by the electrostatic attraction between oppositely charged ions. While the individual ions are charged, the ionic compound itself is electrically neutral. For example, sodium chloride is made up of Na^+ and Cl^- ions but the compound is electrically neutral.

Writing the chemical formula for binary ionic compounds

In section S1.4.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/empirical-formula-and-molecular-formula-id-43572/) you learned about the empirical formula of a compound as the simplest ratio of atoms of each element present in that compound. The chemical formula for ionic compounds is known as a formula unit. It represents the simplest ratio of ions making up the ionic compound and is also an example of an empirical formula. If you look at the formula unit of an ionic compound such as NaCl, this means that there is a 1:1 ratio of Na^+ to Cl^- ions. This does not mean that there is only one of each ion, since ionic compounds are found in a crystal lattice structure consisting of many ions.

You can deduce the formula for binary ionic compounds by considering the number of valence electrons. As ionic compounds are electrically neutral, the ratio of cations to anions is a result of the number of electrons lost from the metal and gained in the non-metal. Let's look at **Figure 2** to illustrate how to deduce the chemical formula for two different ionic compounds.

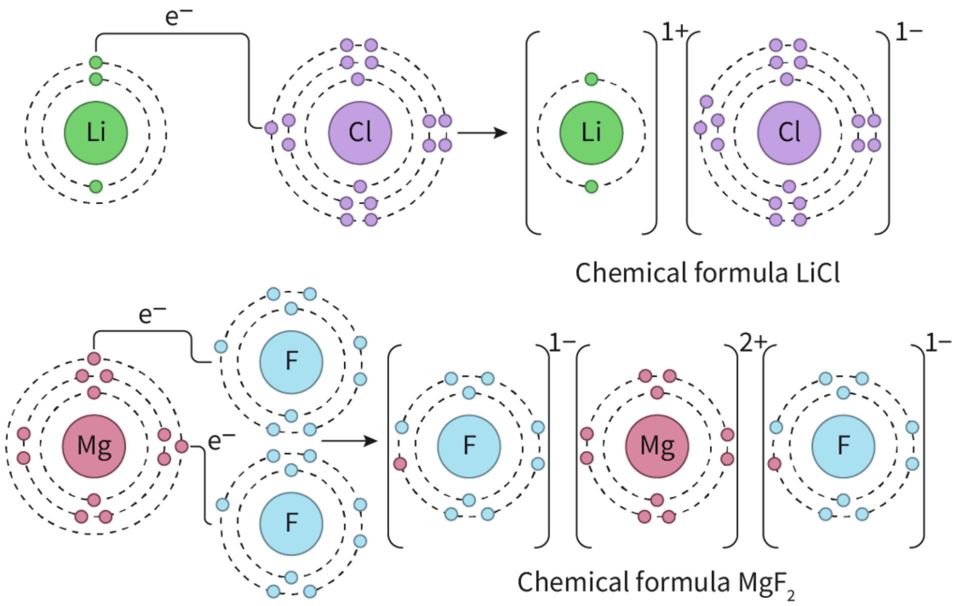


Figure 2. Formation of ionic compounds LiCl and MgF_2.

More information for figure 2

The image is a diagram illustrating the formation of two ionic compounds: LiCl and MgF2. In the formation of LiCl, lithium (Li) is shown with one valence electron, while chlorine (Cl) is shown with seven valence electrons. Lithium donates its one electron to chlorine, forming a Li+ cation and a Cl- anion, which combine to form LiCl.

In the formation of MgF2, magnesium (Mg) is illustrated with two valence electrons, while each fluorine atom (F) is shown with seven valence electrons. Magnesium donates one electron to each fluorine atom, resulting in an Mg2+ cation and two F- anions. They combine in a 1:2 ratio to form MgF2. The diagram uses Lewis dot structures to represent the electron transfers and resulting ions.

[Generated by AI]

In the first example in **Figure 2**, lithium has one valence electron so it will form the Li^+ ion. It can transfer that one valence electron to a chlorine atom to form the Cl^- ion. Since they both have the same (but opposite) charge, the ratio of cations to anions will exist in a 1:1 ratio. Therefore, the chemical formula will be LiCl.

Study skills

Note that when writing the chemical formula for ionic compounds the cation is **always** written first. When you are deducing the chemical formula of an ionic compound, it is necessary to know the charges of the individual ion. These charges are helpful to determine the chemical formula, however, ensure that no charges are shown when writing the chemical formula of ionic compounds. The ionic compound is electrically neutral; it is just composed of charged ions.

In the second example in **Figure 2**, magnesium has two valence electrons so will form the Mg^{2+} ion. It has two electrons available for transfer, however, fluorine only needs to gain one electron from magnesium to have a full valence shell. This means there is another electron available to be transferred to a second fluorine atom. Therefore, there are two F^{-} ions for every one Mg^{2+} ions. The ratio of cations to anions will exist in a 1:2 ratio. Hence, the chemical formula will be MgF_2.

Making connections

The formation of an ionic compound from its elements, where some elements lose electrons and the others gain elections, is an example of a redox reaction. You will explore this type of reaction in subtopic R3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/).

Another way to deduce the formula is to use what is known as a 'swap and drop' approach. Using this method, you would:

- deduce the charge on each ion
- if the charges of the ions are different **swap** the number of charges and **drop** the signs
- the simplest ratio of the swapped charges helps you deduce the chemical formula.

Consider the three examples shown in **Table 1**.

Table 1. Swap and drop method for determining chemical formula of ionic compound

	Sodium fluoride		Sodium sulfide		Magnesium oxide	
Symbol	Na^{+}	F^{-}	Na^{+}	S^{2-}	Mg^{2+}	O
Charge	1+	1−	1+	2−	2+	2
Swap and drop	Charges are the same Write in 1:1 ratio		Charges are diffe rent Swap and drop		Charges are the same 2:2 ratio which can be red to 1:1 ratio	
Ratio	1	1	2	1	1	1
Chemical formula	NaF		Na_2S		MgO	

	Sodium fluoride	Sodium sulfide	Magnesium oxide
Note	When no subscript is written it is assumed to have a value of one		

Naming binary ionic compounds

You have probably heard of some names of ionic compounds in your everyday life. Sodium chloride in table salt, potassium iodide in iodised salt and sodium fluoride in some fluoride containing toothpastes. Do you notice anything about these names? Which ion is named first? Which ion is named second? Do the names of the ion differ from the names of the neutral element?

Nature of Science

Aspect: Science as a shared endeavour

Scientists aim to develop common terminology and conventions, such as naming ionic compounds, to facilitate unambiguous communications globally.

When naming binary ionic compounds (those made up of only two elements), the cation is listed first, followed by the anion. The name of the cation is simply the same as the neutral element. For the anion, the root of the full name is used, but the suffix is changed to '-ide': for example, when a chlorine atom gains an electron it becomes a chlor**ide** ion. The subscript does not play a role in the name of an ionic compound since the compound has a fixed ratio of ions to ensure it is electrically neutral.

The name of the cation always remains the same as the name of the element, but the name of the anion changes to have an '-ide' ending. **Table 2** lists some elements and names of anions.

Table 2. Elements and anion names.

Element name	Anion name	Element name	Anion nar
Hydrogen	Hydride	Oxygen	Oxide
Fluorine	Fluoride	Sulfur	Sulfide
Chlorine	Chloride	Selenium	Selenide
Bromine	Bromide	Nitrogen	Nitride

Element name	Anion name	Element name	Anion nar
Iodine	Iodide	Phosphorus	Phosphic

If the cation is a transition element, roman numerals are used in brackets to denote the charge of the cation when naming. Let's look at a few examples in **Table 3.**

Table 3. Name and chemical formula for a few ionic compounds.

Ions	Name	Chemical formula
Li^{+} and S^{2-}	Lithium sulfide	Li_2S
Cu^{2+} and Cl^{-}	Copper(II) chloride	$CuCl_2$
Fe^{3+} and O^{2-}	Iron(III) oxide	Fe_2O_3

Exercises 1 — 4

Click a question to answer

1. What is the chemical formula for magnesium iodide?

Type here

2. What is the name for CuO?

Type here

3. What is the chemical formula for chromium(III) oxide?

Type here

4. What is the name for Na_3N?

Check answers

Polyatomic ions

Did you know that it is possible to form an ion from more than one atom? Ions can be made up of more than one atom that, overall, has experienced a loss or gain of electrons. These are known as polyatomic ions. These types of ion are made up by atoms held together by covalent bonds but, overall, have experienced a loss or gain of electrons. You will learn more about covalent bonding in section S2.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/). This causes this group of atoms to have a positive or negative charge and can form an ionic bond with an ion of opposite charge.

Table 4 shows some of the most common polyatomic ions that you will encounter in this course.

Table 4. Common examples of polyatomic ions.

Name of polyatomic ion	Overall charge of ion	Form
Ammonium	1+	NH_4
Hydroxide	1−	OH
Nitrate	1−	NO_3
Hydrogencarbonate	1−	HCO
Carbonate	2−	CO_3
Sulfate	2−	SO_4
Phosphate	3−	PO_4

Study skills

Note that the names and formula for polyatomic ions **are not** provided in the DP chemistry data booklet. This means that you must be familiar with those listed in **Table 4** for assessments.

Figure 3 depicts an ionic bond between the ammonium and chloride ions illustrating that this compound would contain both ionic and covalent bonds.

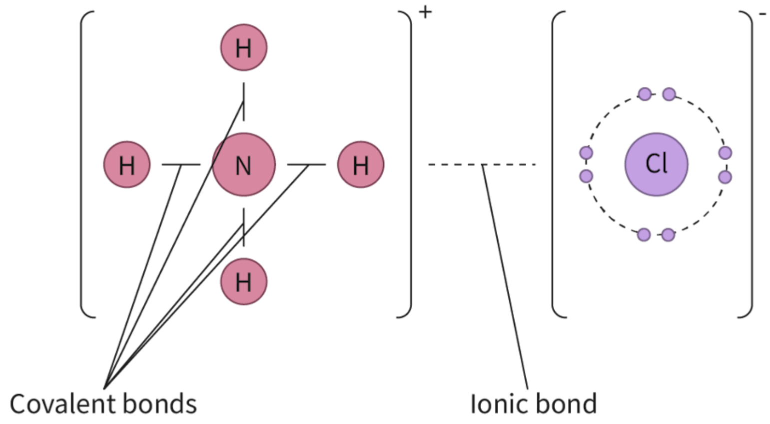


Figure 3. Bonding within ammonium chloride.

More information for figure 3

The diagram illustrates the bonding structure within ammonium chloride. On the left side, there is an ammonium ion (NH_4^+), depicted as a central nitrogen (N) atom bonded to four surrounding hydrogen (H) atoms, each connected by covalent bonds, shown as lines between the atoms. The entire ammonium ion is enclosed within a bracket marked with a positive sign to indicate its overall positive charge.

On the right side, the chloride ion (Cl^-) is shown, represented by a central chlorine (Cl) atom surrounded by dotted lines depicting a shell with seven electrons, symbolizing its negative charge. The chloride ion is also enclosed within brackets marked with a negative sign.

A dotted line between the ammonium and chloride ions signifies the ionic bond that holds these two parts together, illustrating the combination of ionic and covalent bonding within ammonium chloride.

[Generated by AI]

How might the names and formulas of ionic compounds containing polyatomic ions differ from those of binary compounds? When writing the chemical formula, treat the entire polyatomic ion as one unit, using brackets when there is more than one of the same polyatomic ion. When naming an ionic compound containing a polyatomic ion, simply state the name of the polyatomic ion. **Figure 4** shows how you can deduce the chemical formula for the ionic compound ammonium phosphate.

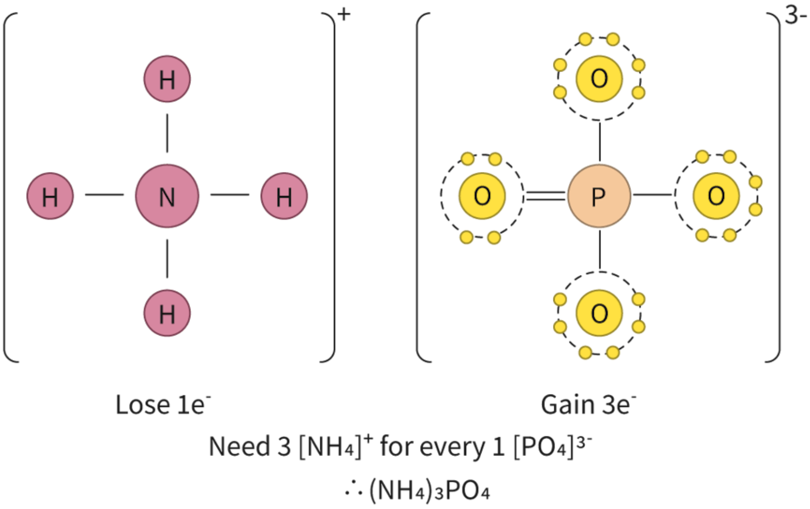


Figure 4. Deducing the chemical formula for ammonium phosphate.

More information for figure 4

The image is a diagram illustrating the chemical relationship between ammonium and phosphate ions to form ammonium phosphate. On the left side, there is a representation of the ammonium ion (NH_4^+), which consists of a central nitrogen (N) atom surrounded by four hydrogen (H) atoms, each connected with a single line, indicating chemical bonds. There is a positive charge symbol near the brackets surrounding the ion, indicating a loss of one electron.

On the right side, the phosphate ion (PO_4^{3-}) is shown with a central phosphorus (P) atom connected to four oxygen (O) atoms with double bonds and enclosed in brackets. Around each oxygen atom, there are dots representing gained electrons, and a negative charge of three is shown outside the brackets.

Below these representations, text reads "Lose 1e^-" near the ammonium ion and "Gain 3e^-" near the phosphate ion. Further text states "Need 3 $[NH_4]^+$ for every 1 $[PO_4]^{3-}$ ∴ $(NH_4)_3PO_4$," indicating the combination of ions to form ammonium phosphate.

[Generated by AI]

We can similarly use the swap and drop method to determine the chemical formula for ionic compounds containing polyatomic ions. Using this method, you would:

- deduce the charge on each ion
- if the charges of the ions are different **swap** the number of charges and **drop** the signs
- the simplest ratio of the swapped charges helps you deduce the chemical formula
- if more than one polyatomic ion is needed in the chemical formula, use brackets around it.

Table 5. Swap and drop method for determining chemical formula of ionic compounds inv polyatomic ions.

	Ammonium chloride		Iron(III) sulfate	
Symbol	NH_4^+	Cl^-	Fe^{3+}	SO_4
Charge	1+	1−	3+	2−
Swap and drop	Charges are the same Write in 1:1 ratio		Charges are differe Swap and drop	
Ratio	1	1	2	3
Chemical formula	NH_4Cl		$Fe_2(SO_4)_3$	

Exercise 5

Click a question to answer

1. What is the chemical formula for magnesium iodide?

Check answers

In the next activity you will use a matching puzzle to reinforce your learning about ionic compounds.

Activity

- **IB Learner profile attribute:** Communicators
- **Approaches to Learning:** Social — Collaborating with a peer to consolidate understanding and to work towards a common goal
- **Time required to complete activity:** 15—20 minutes
- **Activity type:** Group activity (2—3 students)

Download the following Tarsia puzzle.

Tarsia puzzle
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

In groups of 2—3, complete the Tarsia puzzle by placing the sides of the triangles that match one another beside each other. Ensure all members of the group make equal contributions.

S2.1 The ionic model

Structure of ionic compounds

Learning outcomes

By the end of this section you should be able to:

- Describe the three-dimensional structure of an ionic compound.
- Predict the relative strength and an ionic bond by considering lattice enthalpy, ionic radius and charge.

Ionic minerals make up some of the most beautiful naturally occurring solids. Calcium carbonate is an ionic compound that comprises the main component in the shells of marine organisms, eggshells, rocks, pearls, stalagmites and stalactites. What is it about the structural arrangement of ions in the solid that allows for the formation of unique macroscopic structures?

Interactive 1 shows some examples of where calcium carbonate can be found in nature. Scroll through to see some naturally occurring calcium carbonate structures.

Interactive 1. Different Examples of Calcium Carbonate Found in Nature.

More information for interactive 1

An interactive slideshow with five slides presents photographs of naturally occurring calcium carbonate.
Users can switch between slides using arrows on the right and/or left. At the bottom of the page, there are five small circular indicators representing the total number of images in the interactive slideshow. Based on the image viewed, the specific indicator turns blue. The zoom-out arrow at the top right allows the users to view the images in full-screen mode.
Read below to know the various naturally occurring structures with calcium carbonate:
Slide 1: A group of mussel shells.
Slide 2: Several broken eggshells are placed on a brown wooden surface.
Slide 3: A rocky mountain next to a water body. The surface of the land and most of the mountain appears white.
Slide 4: A pearl inside an oyster shell.
Slide 5: Stalagmite and stalactite formations inside a cave.

In section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) you learned how the ionic bond is formed by the electrostatic attraction between oppositely charged ions. Does this attraction only happen between two oppositely charged ions or can it happen between many ions of opposite charge? This electrostatic attraction occurs between many ions in three-dimensions in a predictable arrangement to create a highly ordered solid called a lattice. All ionic compounds exist as three-dimensional lattice structures.

Interactive 2 shows how the lattice structure of lithium fluoride is formed from its elements. Press the ‘next’ button to watch the formation of an ionic lattice of LiF from its elements.

Interactive 2. Formation of an ionic lattice of LiF from its elements.

More information for interactive 2

An interactive animated slideshow displays the formation of a lattice structure of lithium fluoride. Next and previous buttons are on the bottom left allowing navigation between slides.
Slide 1: Fluorine is a gas and has atoms that exist in pairs. A balloon depicts pairs of fluorine atoms. Lithium is a solid metal arranged with lithium atoms.
Slide 2: One lithium atom can transfer its valence electron to one fluorine atom. Two units of Li^+ and F^- are formed after the electron transfer. Li^+ ions are smaller than F^- ions. Each F^- ion is attracted to one F^- ion. Ionic bonds are formed between oppositely charged ions.
Slide 3. The ions in the pairs experience an electrostatic attraction to all ions of opposite charge. Two units of Li^+ ion and F^- ion interact with each other.
Slide 4. All the formed ions self-assemble into a crystal lattice structure with the oppositely charged ions. The ions are arranged alternatively in the lattice.
Slide 5. This process happens with all available ions in three dimensions to form a cubic

Kog

lattice.
A play/pause button is located at the bottom left to control the animation. An option for fullscreen is available at the bottom right.
This interactivity helps users understand the formation of LiF through ionic bonding and the electrostatic attraction between lithium and fluoride ions.

In **Interactive 2**, you can see that the ionic lattice of lithium fluoride is a three-dimensional repeating arrangement of ions, forming a crystalline solid. Notice how the formula of lithium fluoride is LiF, even though the structure of the compound contains many ions. It is important to remember that the chemical formula for ionic compounds is an empirical formula, which represents the simplest ratio of ions that make up that compound. In a three-dimensional structure of LiF, there are an equal number of Li^+ ions as there are F^- ions.

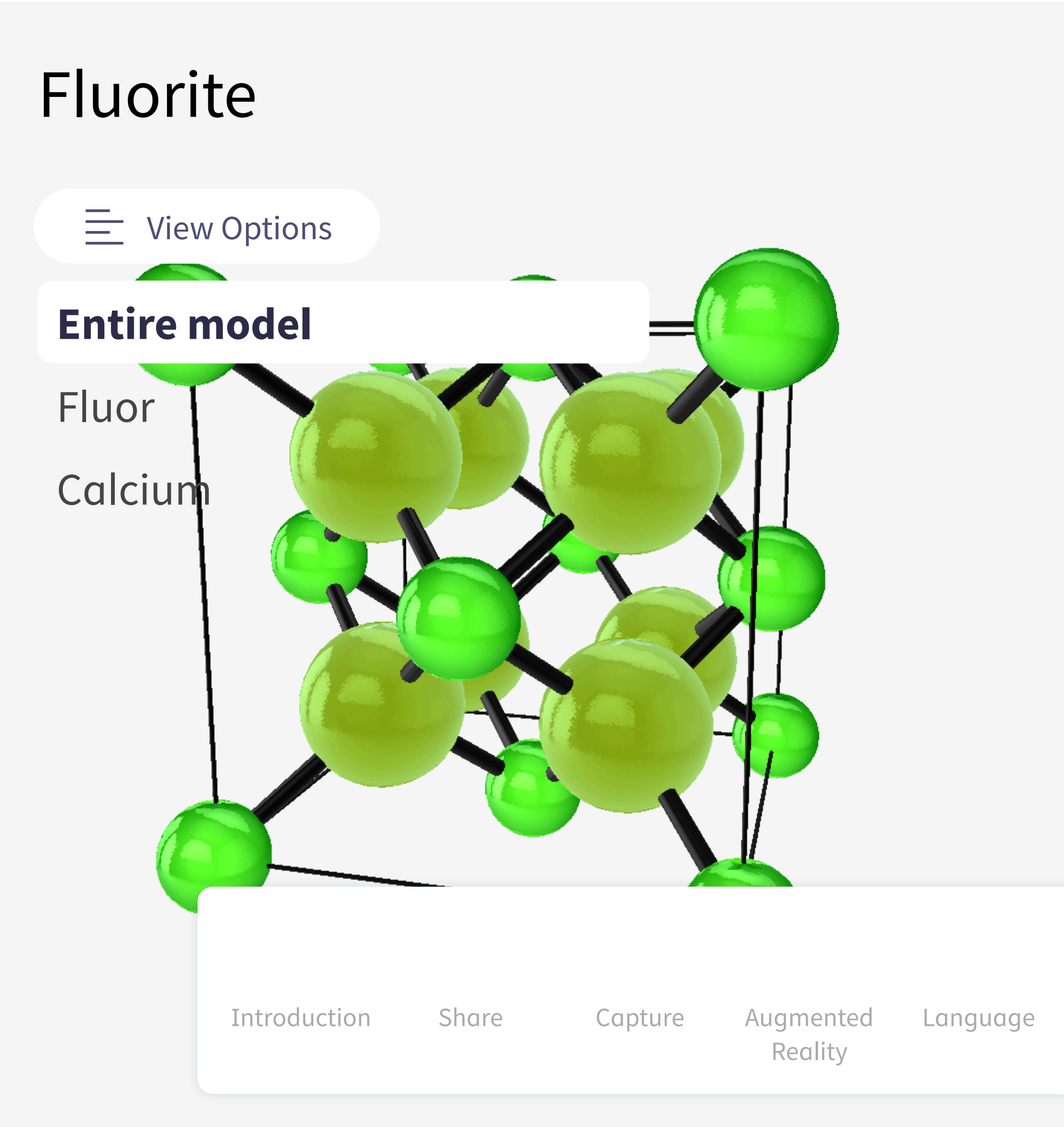
Let's look at two different lattice structures below in **Interactive 3**. Rotate and zoom the graphics to see the lattice structures for halite (NaCl) and fluorite (CaF_2) in more detail.

Interactive 3. Structure of Halite ($NaCl$).

More information for interactive 3

An interactive three-dimensional model depicts a ball-and-stick model of halite ($NaCl$). This model features a cubic structure where sodium ions occupy the corners and the centers of each face of the cube and chlorine ions occupy the center of each edge and the center of the cube.
Users can rotate the model to view it from different angles and zoom in or out using the mouse scroll. In the model, the sodium and chlorine ions are displayed as smaller and larger spheres, respectively.
A panel on the left provides view options, including the entire model, chlorine, and sodium. Selecting chlorine or sodium highlights their arrangement within the model.
On the right side, option for changing the background color and option for zooming the interactive are given.
At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction option displays the following details:
Chemical name: Sodium chloride
General formula: $NaCl$
Molar mass: 58.44 g/mol

Mohs scale hardness: 2
Crystal system: Cubic
Unit cell: Face-centered cubic lattice, with Na^+ and Cl^- in alternating positions.
Crystal habit: Predominantly forms cubic crystals.
Description: Pure sodium chloride, also known as halite in its natural form, is a colorless crystalline solid with a salty taste. It readily dissolves in water to produce brine, an electrically conductive solution.
Applications: Sodium chloride is used in the food industry as a seasoning and preservative and in the chemical industry as a raw material for producing various sodium and chlorine compounds.
This interactivity helps users understand the position of sodium and chloride ions in halite and their ratio in the crystal structure.



Interactive 3. Structure of Fluorite (CaF_2).

More information for interactive 3

An interactive three-dimensional model depicts a ball-and-stick model of fluorite (CaF_2). This model features a cubic structure where calcium ions occupy the corners and the centers of each face of the cube and fluoride ions occupy the tetrahedral voids within the cubic lattice. Each fluoride ion is connected to four calcium ions.
Users can rotate the model to view it from different angles and zoom in or out using the mouse scroll. In the model, the calcium and fluoride ions are displayed as smaller and larger spheres, respectively.
A panel on the left provides view options, including the entire model, fluor, and calcium. Selecting fluor or calcium highlights their arrangement within the model.
On the right side, options for changing the background color and option for zooming the interactive are given.
At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction option displays the following details:
Chemical name: Calcium fluoride
Molecular formula: CaF_2
Molar mass: 78.07 g/mol
Mohs scale hardness: 4
Crystal system: Face-centered cubic lattice of Ca^{2+} ions, with F^- ions occupying the tetrahedral cavities.
Crystal habit: Typically forms cubic crystals, occasionally exhibiting octahedral shapes.
Description: Pure calcium fluoride is a colorless crystalline substance that is insoluble in water. In nature, the presence of trace amounts of rare earth elements gives fluorite a variety of colors, ranging from blue, yellow, and white to purple and even black.
Applications: Fluorite (calcium fluoride) is used in the production of hydrofluoric acid. It is also used in metallurgy and the glass industry. Additionally, because of its optical properties, calcium fluoride is commonly used as a material for manufacturing lenses and other optical components.
This interactivity helps users understand the position of calcium and fluoride ions in fluorite and their ratio in the crystal structure.

In **Interactive 3**, the lattice structure of two minerals, halite and fluorite, are shown. Halite is the ionic compound NaCl. Notice how there is an equal ratio of sodium ions to chloride ions in the lattice structure. Fluorite is the ionic compound CaF_2. Notice how there are two times as many fluoride ions as there are calcium ions in the lattice structure.

International Mindedness

The term lattice is used in science to denote a regular repeating three-dimensional structure of ions, atoms, or molecules, but it is also used in other contexts. Lattice multiplication is a specific technique of multiplying numbers using a grid. This method of multiplication has been used historically by many different cultures. For example, there is evidence of this in Arabic, Chinese and European mathematics.

Strength of the ionic bond

The strength of an ionic bond varies between different compounds. Lattice enthalpy is a measure of the strength of the electrostatic attraction between ions in an ionic compound. You will learn more about lattice enthalpy in section R1.2.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/born-haber-cycle-hl-id-43579/). Values of lattice enthalpy can be found in section 16 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lattice-enthalpies-at-29815-k-experimental-values-id-45118/) of the DP chemistry data booklet.

- The larger the lattice enthalpy the stronger the ionic bond.
- The lower the lattice enthalpy the weaker the ionic bond.

Table 1 shows a summary of the lattice enthalpies for group 1 and 2 chlorides. Looking at this data, can you think of any factors that would influence the strength of the ionic bond?

Table 1. Lattice enthalpies and cationic radius of group 1 and

Group 1 chlorides	Radius of cation (10^{-12} m)	Lattice enthalpy (kJ mol^{-1})	Group 2 chlorides	Radiu cation (m)
LiCl	76	864	$BeCl_2$	45
NaCl	102	790	$MgCl_2$	72
KCl	138	720	$CaCl_2$	10(
RbCl	152	695	$SrCl_2$	118
CsCl	167	670	$BaCl_2$	135

The box below mentions a different, but similar concept to lattice enthalpy that you will encounter later on in the course.

Kog

Making connections

In section R1.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-breaking-and-forming-id-43473/) you learn about bond enthalpy. Bond enthalpy is a similar concept to lattice enthalpy but a measure of the strength of covalent bonds. Both are quantitative, energetic measurements. You learn more about quantitative, energetic measurements known as enthalpy in section R1.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpy-and-heat-transfers-id-43576/).

Radius of the ions

Recall from section S1.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) that the atom is mostly empty space. The radius of an atom or ion is determined dependent on the number of occupied energy levels. The more occupied energy levels there are, the larger the radius of an ion. Do you think larger or smaller ions would have a stronger attractive force with one another?

Looking at **Table 1**, you can see, for group 1 chlorides, as the radius of the cation increases, the lattice enthalpy decreases. This means that the larger the radius of the ion, the weaker the ionic bond. This is because of the greater distance between the two ions as the radius increases, weakening the electrostatic attraction.

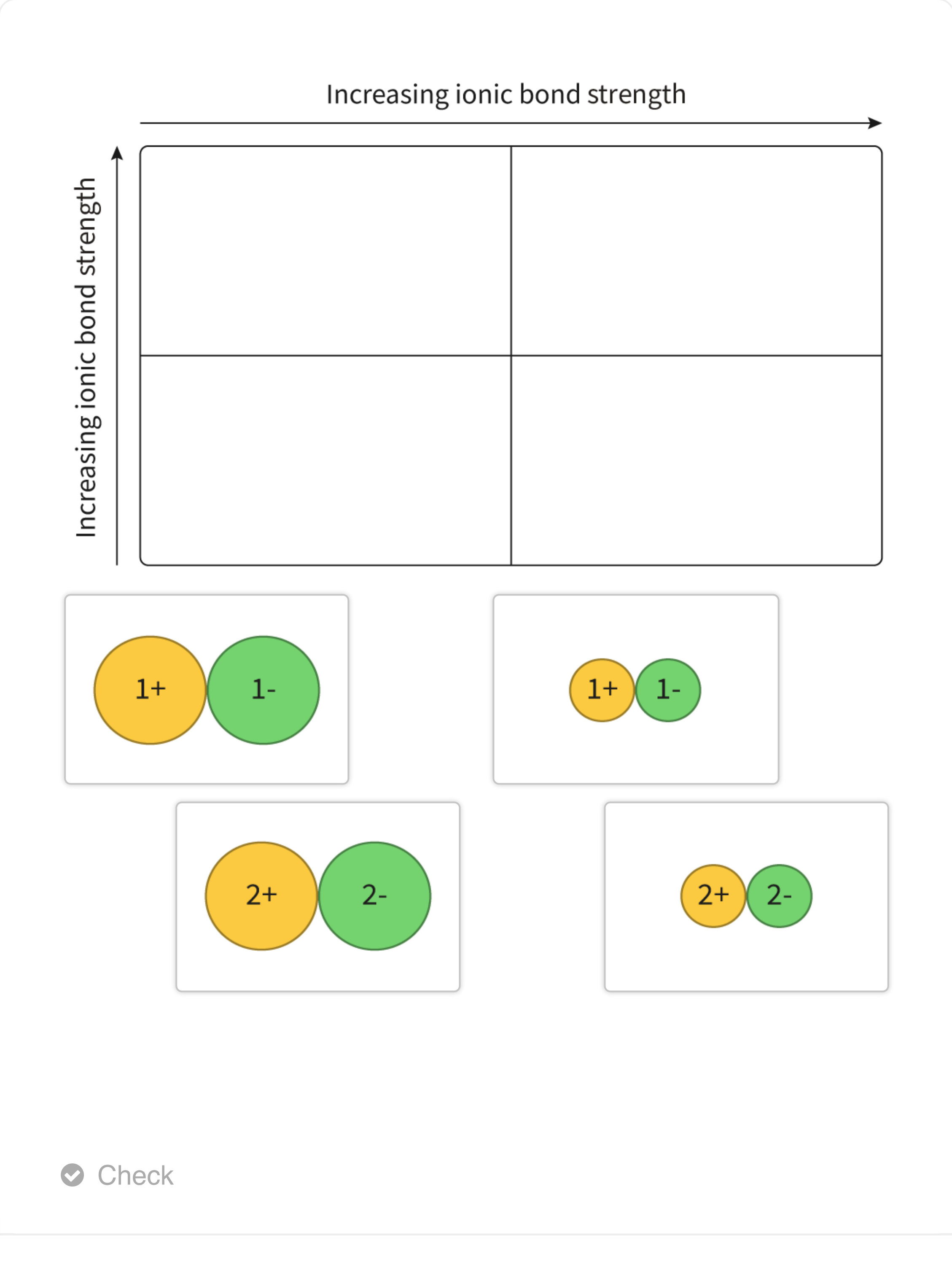
Charge of the ions

Recall that the ionic bond is the electrostatic attraction between oppositely charged ions. How do you think the charge on cations and anions would affect the attractive forces between them?

Looking at **Table 1**, you can see that group 2 chlorides have considerably higher lattice enthalpies than group 1 chlorides. This is due to the greater charge on the cation. Group 2 cations would have a 2+ charge whereas group 1 ions have a 1+ charge. This greater charge results in a stronger electrostatic attraction and a stronger ionic bond.

In summary, the smaller the radius and the greater the charge of the ions, the stronger the ionic bond. The larger the radius and the lower the charge of the ions, the weaker the ionic bond.

In **Interactive 4,** drag and drop pairs of ions according to their ionic bond strength.



Interactive 4. Pairs of Ions According to Their Ionic Bond Strength.

More information for interactive 4

An interactive 2×2 grid challenges users to analyze and arrange different ion pairs based on their ionic bond strength. The grid is guided by directional arrows: ionic bond strength increases from left to right (→) and from bottom to top (↑), encouraging learners to make comparisons based on both ion size and ionic charge.
Users must drag and drop four distinct ion pair models into the appropriate grid cells.

The models provided are as follows:
A large yellow sphere with a 1+ charge is in contact with a large green sphere with a 1- charge.
A large yellow sphere with a 2+ charge is in contact with a large green sphere with a 2- charge.
A small yellow sphere with a 1+ charge is in contact with a small green sphere with a 1- charge.
A small yellow sphere with a 2+ charge is in contact with a small green sphere with a 2- charge.
This activity visually reinforces the concept that ionic bond strength is influenced by ionic size and charge, with stronger bonds forming between smaller ions and ions with higher charges.
Read below for the solution:
Column 1, Row 1: A small yellow sphere with a 1+ charge is in contact with a small green sphere with a 1- charge.
Column 2, Row 1: A small yellow sphere with a 2+ charge is in contact with a small green sphere with a 2- charge.
Column 1, Row 2: A large yellow sphere with a 1+ charge is in contact with a large green sphere with a 1- charge
Column 2, Row 2: A large yellow sphere with a 2+ charge is in contact with a large green sphere with a 2- charge.

The box below links the two factors that influence the strength of the ionic bond to electrostatic forces, a concept studied in physics.

Making connections

The electrostatic interaction between two charged particles is an electrostatic **force**, a concept studied in physics. Coulomb's law mathematically describes the electrostatic force between two charged particles:

$$F = k\frac{q_1q_2}{r^2}$$

Where F represents the electrostatic force, k is Coulomb's constant, q_1 and q_2 are the charges of each of the charged particles and r is the distance between the two charged particles. By looking at this equation, it is easy to see that a greater charge and shorter distance will result in a greater force. These were the two factors considered before for the strength of the ionic bond.

For this activity you will investigate the structure of ionic compound using models.

Kog

Activity

- **IB Learner profile attribute:** Thinkers
- **Approaches to Learning:**
  - Thinking — Designing models
  - Research — Evaluating information sources
- **Time required to complete activity**: 30 minutes
- **Activity type:** Pairs, followed by whole class discussion

Make an ionic model!

What you need:

- Plasticine/modelling clay
- Straws/toothpicks

Then follow these instructions:

1. In pairs, create model representing the ionic lattice of:
   (a) Sodium chloride
   (b) Magnesium chloride
2. Discuss with your partner which ionic compound you would expect to have stronger ionic bonds and why.
3. After the creation of your models, research the structure for each of these.
4. As a class, consider the following questions.
   (a) What parts of your structure were similar to the researched structure?
   (b) What parts of your structure were different from the researched structure?
   (c) How would you assess the credibility of the source you obtained information from?
   (d) What part of this task did you find simple?
   (e) What part of this task did you find difficult? Why?

Physical properties of ionic compounds

Learning outcomes

By the end of this section you should be able to explain the physical properties of ionic compounds in connection to its structure and bonding for the following:

- melting point
- volatility
- solubility in water
- electrical conductivity
- brittleness.

Ionic compounds are commonly found as salts, minerals and ceramics. Society typically uses the term salt to refer to sodium chloride, however, in chemistry this term is used to specify an ionic compound. While you might be most familiar with sodium chloride (table salt), salts exist in many different colours depending on the nature of the cation, anion or even the solvent within the lattice structure. Scroll through the images in **Interactive 1** to see some different examples of salts you may encounter in the course.

Interactive 1. Examples of Different Salts.

More information for interactive 1

Thinking about table salt (NaCl), an ionic compound you are likely familiar with from everyday life, what are some properties you know about this salt? If you add it to a pot of water, will it dissolve? Does it melt easily? Does it break easily into small pieces? Can it conduct electricity in its solid state? What about as a solution in water? Think about your experiences with table salt as we look more closely at the physical properties of ionic compounds.

Melting points

Think about all the examples of ionic compounds we have seen so far. What state of matter do you notice most of them have been in? Nearly every example of ionic compound we have looked at has been in its solid state. In section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) you learned what change of state occurs during the process of melting. A melting point is the temperature at which a solid will undergo a change of state to a liquid. Would you expect substances that are solid at room temperature to have high or low melting points?

Melting an ionic compound to form a liquid involves breaking the electrostatic attractive forces between the oppositely charged ions in the lattice. Due to the strength and number of the electrostatic attractive forces between ions, a lot of energy is required to melt ionic compounds. Therefore, ionic compounds have high melting points. **Table 1** summarises some of the melting points for group 1 and 2 chlorides. Notice that these values are all quite high, much higher than can be achieved with an oven in your home!

Table 1. Melting points for group 1 and 2 chlorides.

Group 1 chlorides	Melting point (°C)	Group 2 chlorides	Melting point (
$LiCl$	605	$BeCl_2$	415
$NaCl$	801	$MgCl_2$	714
KCl	770	$CaCl_2$	772
$RbCl$	718	$SrCl_2$	874
$CsCl$	645	$BaCl_2$	962

The box below looks at a group of ionic compounds that have lower melting points than those typically referenced when discussing ionic compounds.

International Mindedness

Ionic liquids are a group of ionic compounds with significantly lower melting points (<100 °C). Ionic liquids are unique in that they can dissolve a large range of different solutes. Globally, many current solvents used in industry are flammable and give off volatile compounds that are harmful to the surrounding environment. Using ionic liquids as solvents reduces these concerns. Ionic liquids form an active area of research in Green Chemistry, an area of chemistry focused on the minimisation and elimination of hazardous substances. Visit this source (https://www.electrochem.org/dl/interface/spr/spr07/spr07_p38.pdf) if you are interested in learning more about ionic liquids.

Volatility

The term volatility describes the ease at which a substance vaporises or becomes a gas. In section S1.1.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-id-43066/) you learned which changes of state are involved in volatility. Volatility would be the ease at which a substance evaporates, boils or sublimates. A substance with a high volatility easily becomes a gas. You have likely encountered volatile substances as the 'smelly' substances in vinegar, perfume and paints. A substance with a low volatility does not become a gas easily. As ionic compounds have high melting points and do not easily vaporise they have a low volatility.

There are, however, substances known as smelling salts. These salts are usually made of ammonium carbonate and were traditionally used to relieve faintness due to the odour given off. Recall what you learned about the strength of ionic bonds and lattice enthalpy in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). How do you think the lattice enthalpy of smelling salts compares to that of typical salts?

Solubility

Have you ever noticed that when you shake table salt into water it dissolves? From section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/), what is the name of the homogeneous mixture formed when table salt dissolves in water? When sodium chloride in table salt dissolves in water an aqueous solution is formed. Sodium chloride dissolves in water due to the strong interactions sodium and chloride ions have with water compared to between the oppositely charged ions in the lattice. These strong interactions allow water to surround individual ions, separating ions from their fixed positions in the lattice. **Figure 1** shows how a solution is formed from solid sodium chloride when it encounters water.

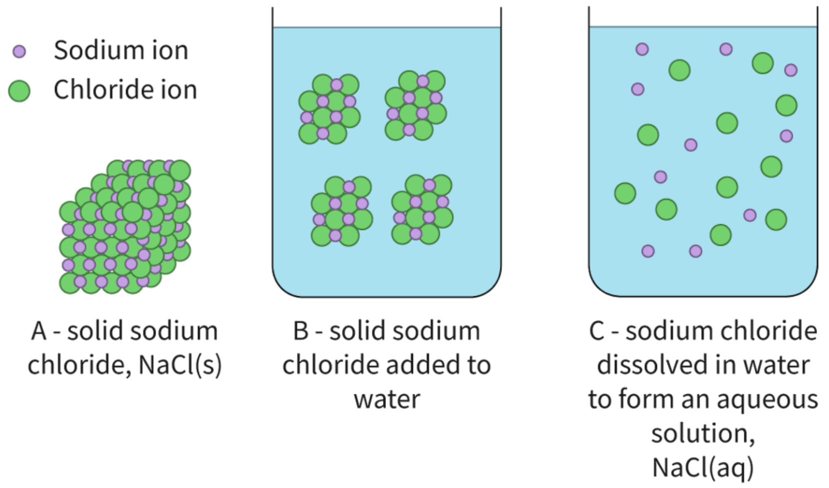


Figure 1. Solid sodium chloride dissolving in water.

More information for figure 1

We can represent the physical change occurring during the formation of an aqueous solution for sodium chloride using state symbols as:

$$NaCl(s) \rightarrow NaCl(aq)$$

Since individual ions are surrounded by water in the aqueous solution of an ionic compound, we can also write the physical change occurring as:

$$NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$

Many, but not all, ionic compounds are soluble in water. The solubility of an ionic compound depends on the strength of the attraction between the ions and water along with the strength of the ionic bond. An ionic compound is insoluble in water when it does not form a homogeneous mixture when added to water.

If you try to shake table salt onto cooking oil, you might notice it does not dissolve. This is because there are no strong interactions between cooking oil and sodium and chloride ions. This makes sodium chloride insoluble in oil. What type of mixture is formed when an ionic compound is insoluble in a solvent?

Theory of Knowledge

Nacre, also known as mother of pearl, is produced on the inner shell of some oysters when a piece of foreign organic material, such as a parasite, enters the shell, or if the oyster's tissue is damaged by a predator. Nacre is composed of layers of calcium carbonate held together by organic material, producing a strong and beautifully iridescent substance highly valued in the art and jewellery industries. Historically, pearl oysters were collected by hand, opened and inspected for pearls — a very tedious practice as finding pearls is very rare and also wasteful, greatly depleting the pearl oyster population. Today, 95% of pearls available are 'cultured' by placing a small bead inside an oyster's shell and allowing the oyster to cover the bead with layers of nacre. The resulting pearl is produced faster and with a more consistent shape. This process, developed in 1956, also allowed the pearl oyster population to regenerate to a level that they no longer face extinction. Cultured pearls and natural pearls are almost identical in appearance — the only way to be absolutely certain is to examine the pearl by X-ray. Does the ability to replicate a natural process, producing a nearly identical product, reduce the beauty or value of nacre?

Conductivity

The term electrical conductivity refers to the ability of charged particles to move through a region of space. Recall what you know about ionic compounds. What charged particles are involved? Are these charged particles able to move? Can you think of situations where these charged particles can move?

Solid ionic compounds have cations and anions as charged particles that are fixed in the crystal lattice. This lattice structure in the solid state prevents movement of the charged ions. Therefore, solid ionic compounds are not electrically conductive.

When an ionic compound melts to form a molten liquid, the ionic bonds holding oppositely charged ions begin to break. This allows cations and anions to move throughout the liquid. Since there is motion of charged particles, molten ionic compounds are electrically conductive.

When an ionic compound dissolves in water to form an aqueous solution, ions are surrounded by water. Since ions are no longer fixed in positions (as they were in the solid lattice) they are free to move throughout the solution. As there is motion of charged particles, aqueous ionic compounds are electrically conductive. **Figure 2** illustrates the conductivity for ionic compounds in various states using an electrical circuit.

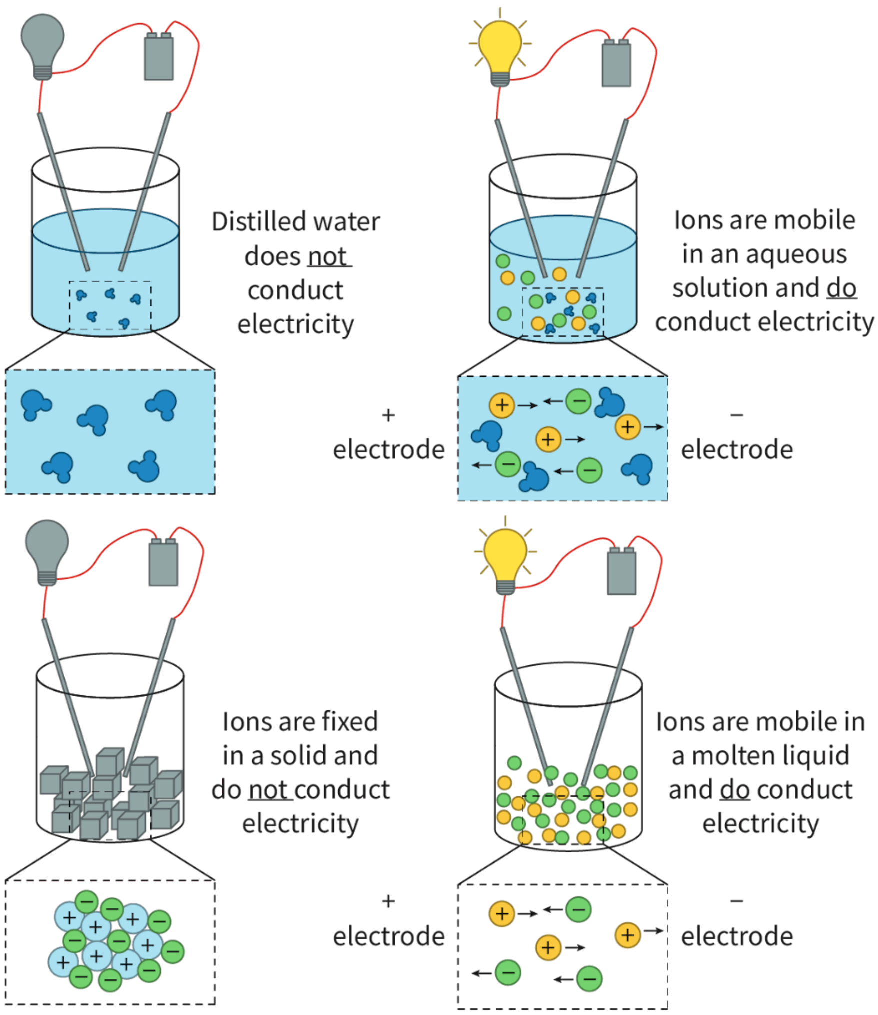
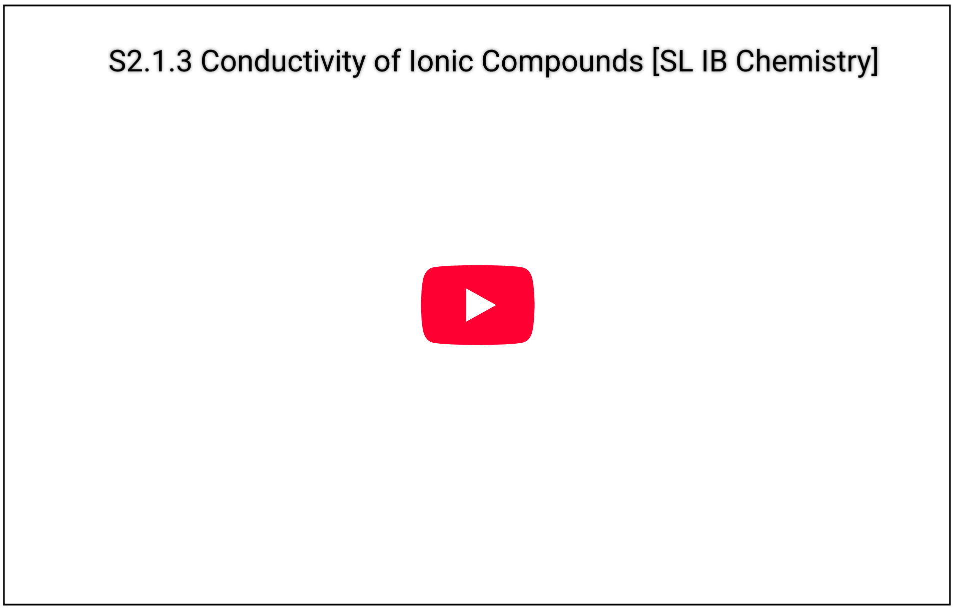


Figure 2. Electrical circuit showing when ionic compounds are electrically conductive.

More information for figure 2

Watch **Video 1** to help you visualise the conductivity of ionic compounds in the states of matter listed above.



Video 1. Visualise the conductivity of ionic compounds in the solid, molten and aqueous states.

The box below highlights common misconceptions of students when it comes to the conductivity of ionic compounds.

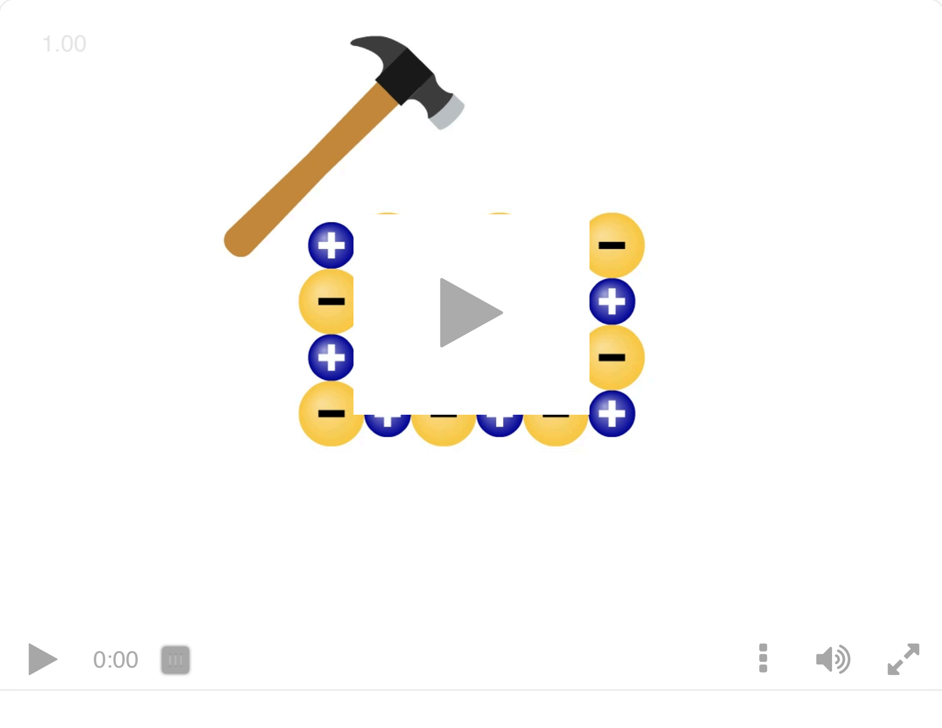
Study skills

Conductivity refers to the mobility of charged particles. It is very common for students to misinterpret conductivity to be exclusively due to electrons, however, in the case of aqueous ionic compounds, the conductivity of the solutions is from the mobility of the ions in solutions, **not** electrons.

Brittleness

Have you ever noticed if you have a large piece of table salt it can easily be ground into smaller pieces? This happens because sodium chloride is brittle. This is the final characteristic property of ionic compounds. While ionic bonds are strong, ionic substances fracture easily. These properties can be explained by taking a close look at the lattice structure as shown in **Interactive 2**.

When a stress is applied (caused by dropping or hitting with a hard object), layers within the lattice shift in response to that stress. As layers shift, ions become adjacent to like charges. These like charges experience an electrostatic repulsion and layers separate from one another, which forces the lattice to break into pieces. This is why ionic compounds, like sodium chloride in table salt, are brittle.



Interactive 2. Ionic Lattice Shifting in Response to an Applied Stress.

More information for interactive 2

The box below provides some examples of methods to collect qualitative and quantitative data for the properties of ionic compounds discussed in this section.

Practical skills

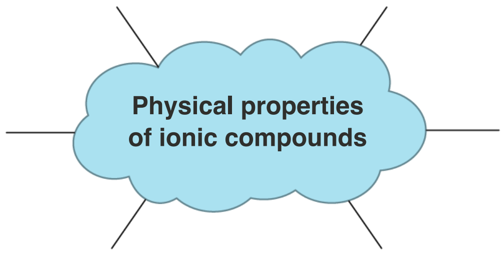
Inquiry 2: Collecting and processing data — Collecting data

There are qualitative and quantitative methods of collecting data to study the physical properties of ionic compounds. Below are examples of a few ways you could collect qualitative and quantitative data to study some of these physical properties.

	Qualitative data	Quantitative data
Melting point	Observe whether or not the substance melts when placed into the flame of a Bunsen burner.	Observe what temperature the substa melts at using a melting point appara
Solubility in water	Observe whether a small amount of the solute forms a homogeneous mixture when added to water.	Use a conductivity meter to measure h conductivity of the solution compares t water. If the ionic compound is colour spectrophotometer can also be usec measure the solubility.

	Qualitative data	Quantitative data
Electrical conductivity	Observe whether a lightbulb of a simple electrical circuit lights up.	Use a conductivity meter to measu conductivity of solution.
Brittleness	Observe (macroscopically with eyes or microscopically with microscope) the ease at which cracks and fractures form on the crystal's surface.	Use a structures and materials teste measures the force at which the mat fractures.

Drag and drop the boxes in **Interactive 3** to see whether you can recall all the physical properties of ionic compounds.



conducts electricity in solid state	low volatility	low melting points	none are soluble in water
conducts electricity in molten state	high volatility	high melting points	some are soluble in water
conducts electricity in aqueous state	brittle	flexible	all are soluble in water

Check

Interactive 3. Summary of the Physical Properties of Ionic Compounds.

More information for interactive 3

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

According to the USGS (https://www.usgs.gov/faqs/why-ocean-salty#:~:text=The%20concentration%20of%20salt%20in,be%20about%20120%20million%20tons.), there are about 120 000 000 tons of salt in a cubic mile of seawater (which is equivalent to 1.09×10^{11} kg in 4.2 km^3 of seawater) and more than 90% of this salt is sodium chloride, an ionic compound. The world's oceans are a vast source of biodiversity but they are at risk from a variety of threats such as plastic pollution and overfishing.

8 June is World Oceans Day (https://www.un.org/en/observances/oceans-day) which aims to increase the public's perception of the threats posed to the world's oceans. What can you do at your school or in your local community to raise awareness of World Oceans Day?

In the following activity you will check your understanding of the physical properties of the ionic compounds covered in this section.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning;** Thinking skills — Combining different ideas in order to create new understanding
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity for steps 1—5, pairs for step 6

1. Looking at the chemical formula for the substances below. Identify which ones you suspect are ionic. Explain how you made your selection.
   (a) $MgCl_2$
   (b) CO_2
   (c) NH_4Br
   (d) Na
2. Research the melting point of each of the substances above. Comment on how you might use the melting point to confirm or refute your selection above.
3. Explain why ionic compounds are brittle using a diagram to support your answer.

4. Sketch a diagram showing what you think happens at the atomic scale when sodium chloride dissolves in water.
5. Create a Venn diagram comparing solid ionic compounds to aqueous ionic compounds. Include structure, properties and diagrams.
6. Compare your work with a peer. Highlight three areas your work had in common. Highlight three areas where your work differed. Discuss these differences with your peer.

S2.1 The ionic model

Physical properties of ionic compounds

Learning outcomes

By the end of this section you should be able to explain the physical properties of ionic compounds in connection to its structure and bonding for the following:

- melting point
- volatility
- solubility in water
- electrical conductivity
- brittleness.

Ionic compounds are commonly found as salts, minerals and ceramics. Society typically uses the term salt to refer to sodium chloride, however, in chemistry this term is used to specify an ionic compound. While you might be most familiar with sodium chloride (table salt), salts exist in many different colours depending on the nature of the cation, anion or even the solvent within the lattice structure. Scroll through the images in **Interactive 1** to see some different examples of salts you may encounter in the course.

Interactive 1. Examples of Different Salts.

More information for interactive 1

Thinking about table salt ($NaCl$), an ionic compound you are likely familiar with from everyday life, what are some properties you know about this salt? If you add it to a pot of water, will it dissolve? Does it melt easily? Does it break easily into small pieces? Can it conduct electricity in its solid state? What about as a solution in water? Think about your experiences with table salt as we look more closely at the physical properties of ionic compounds.

Melting points

Think about all the examples of ionic compounds we have seen so far. What state of matter do you notice most of them have been in? Nearly every example of ionic compound we have looked at has been in its solid state. In section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) you learned what change of state occurs during the process of melting. A melting point is the temperature at which a solid will undergo a change of state to a liquid. Would you expect substances that are solid at room temperature to have high or low melting points?

Melting an ionic compound to form a liquid involves breaking the electrostatic attractive forces between the oppositely charged ions in the lattice. Due to the strength and number of the electrostatic attractive forces between ions, a lot of energy is required to melt ionic compounds. Therefore, ionic compounds have high melting points. **Table 1** summarises some of the melting points for group 1 and 2 chlorides. Notice that these values are all quite high, much higher than can be achieved with an oven in your home!

Table 1. Melting points for group 1 and 2 chlorides.

Group 1 chlorides	Melting point (°C)	Group 2 chlorides	Melting point (
LiCl	605	$BeCl_2$	415
NaCl	801	$MgCl_2$	714
KCl	770	$CaCl_2$	772
RbCl	718	$SrCl_2$	874
CsCl	645	$BaCl_2$	962

The box below looks at a group of ionic compounds that have lower melting points than those typically referenced when discussing ionic compounds.

International Mindedness

Ionic liquids are a group of ionic compounds with significantly lower melting points (<100 °C). Ionic liquids are unique in that they can dissolve a large range of different solutes. Globally, many current solvents used in industry are flammable and give off volatile compounds that are harmful to the surrounding environment. Using ionic liquids as solvents reduces these concerns. Ionic liquids form an active area of research in Green Chemistry, an area of chemistry focused on the minimisation and elimination of hazardous substances. Visit this source (https://www.electrochem.org/dl/interface/spr/spr07/spr07_p38.pdf) if you are interested in learning more about ionic liquids.

Volatility

The term volatility describes the ease at which a substance vaporises or becomes a gas. In section S1.1.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-id-43066/) you learned which changes of state are involved in volatility. Volatility would be the ease at which a substance evaporates, boils or sublimates. A substance with a high volatility easily becomes a gas. You have likely encountered volatile substances as the 'smelly' substances in vinegar, perfume and paints. A substance with a low volatility does not become a gas easily. As ionic compounds have high melting points and do not easily vaporise they have a low volatility.

There are, however, substances known as smelling salts. These salts are usually made of ammonium carbonate and were traditionally used to relieve faintness due to the odour given off. Recall what you learned about the strength of ionic bonds and lattice enthalpy in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). How do you think the lattice enthalpy of smelling salts compares to that of typical salts?

Solubility

Have you ever noticed that when you shake table salt into water it dissolves? From section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/), what is the name of the homogeneous mixture formed when table salt dissolves in water? When sodium chloride in table salt dissolves in water an aqueous solution is formed. Sodium chloride dissolves in water due to the strong interactions sodium and chloride ions have with water compared to between the oppositely charged ions in the lattice. These strong interactions allow water to surround individual ions, separating ions from their fixed positions in the lattice. **Figure 1** shows how a solution is formed from solid sodium chloride when it encounters water.

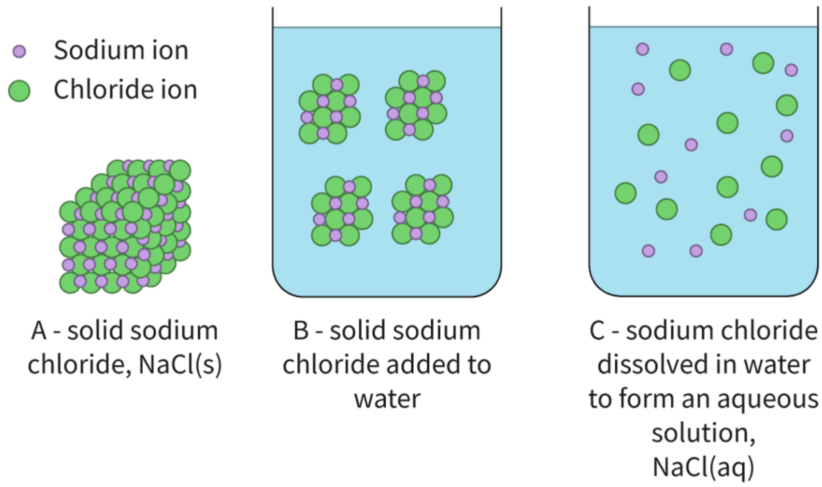


Figure 1. Solid sodium chloride dissolving in water.

More information for figure 1

We can represent the physical change occurring during the formation of an aqueous solution for sodium chloride using state symbols as:

$$NaCl(s) \rightarrow NaCl(aq)$$

Since individual ions are surrounded by water in the aqueous solution of an ionic compound, we can also write the physical change occurring as:

$$NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$

Many, but not all, ionic compounds are soluble in water. The solubility of an ionic compound depends on the strength of the attraction between the ions and water along with the strength of the ionic bond. An ionic compound is insoluble in water when it does not form a homogeneous mixture when added to water.

If you try to shake table salt onto cooking oil, you might notice it does not dissolve. This is because there are no strong interactions between cooking oil and sodium and chloride ions. This makes sodium chloride insoluble in oil. What type of mixture is formed when an ionic compound is insoluble in a solvent?

Theory of Knowledge

Nacre, also known as mother of pearl, is produced on the inner shell of some oysters when a piece of foreign organic material, such as a parasite, enters the shell, or if the oyster's tissue is damaged by a predator. Nacre is composed of layers of calcium carbonate held together by organic material, producing a strong and beautifully iridescent substance highly valued in the art and jewellery industries. Historically, pearl oysters were collected by hand, opened and inspected for pearls — a very tedious practice as finding pearls is very rare and also wasteful, greatly depleting the pearl oyster population. Today, 95% of pearls available are 'cultured' by placing a small bead inside an oyster's shell and allowing the oyster to cover the bead with layers of nacre. The resulting pearl is produced faster and with a more consistent shape. This process, developed in 1956, also allowed the pearl oyster population to regenerate to a level that they no longer face extinction. Cultured pearls and natural pearls are almost identical in appearance — the only way to be absolutely certain is to examine the pearl by X-ray. Does the ability to replicate a natural process, producing a nearly identical product, reduce the beauty or value of nacre?

Conductivity

The term electrical conductivity refers to the ability of charged particles to move through a region of space. Recall what you know about ionic compounds. What charged particles are involved? Are these charged particles able to move? Can you think of situations where these charged particles can move?

Solid ionic compounds have cations and anions as charged particles that are fixed in the crystal lattice. This lattice structure in the solid state prevents movement of the charged ions. Therefore, solid ionic compounds are not electrically conductive.

When an ionic compound melts to form a molten liquid, the ionic bonds holding oppositely charged ions begin to break. This allows cations and anions to move throughout the liquid. Since there is motion of charged particles, molten ionic compounds are electrically conductive.

When an ionic compound dissolves in water to form an aqueous solution, ions are surrounded by water. Since ions are no longer fixed in positions (as they were in the solid lattice) they are free to move throughout the solution. As there is motion of charged particles, aqueous ionic compounds are electrically conductive. **Figure 2** illustrates the conductivity for ionic compounds in various states using an electrical circuit.

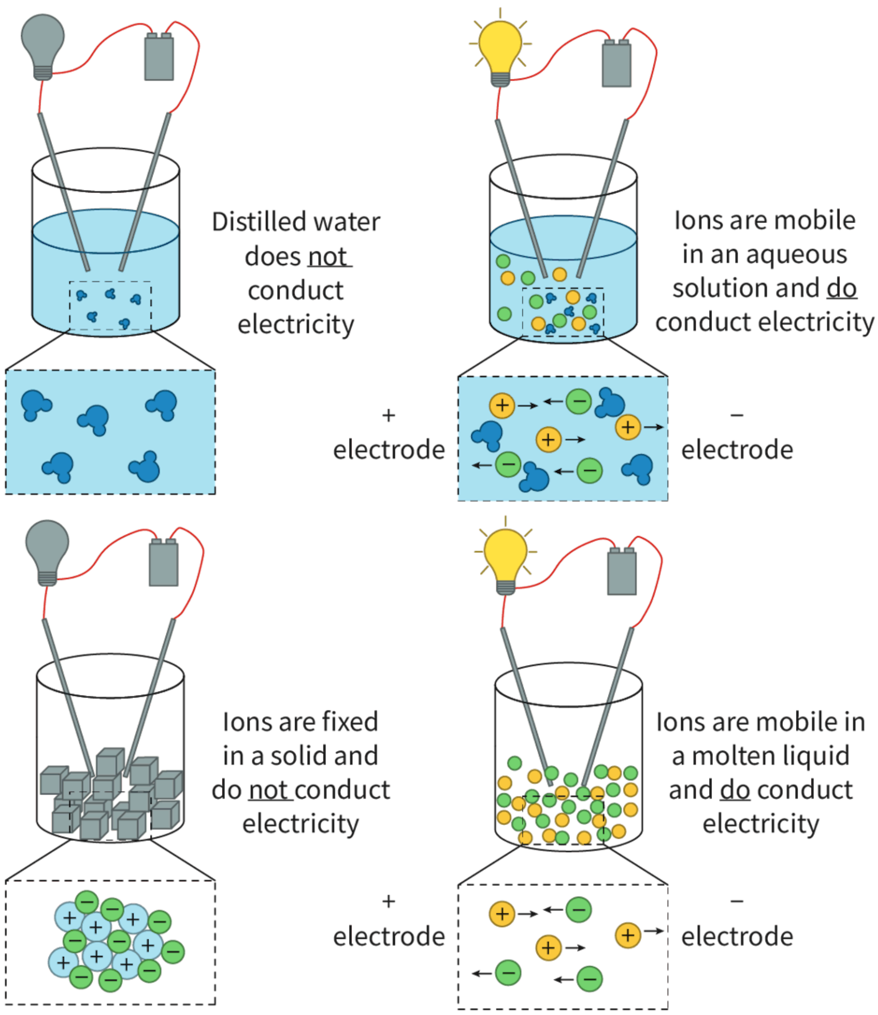
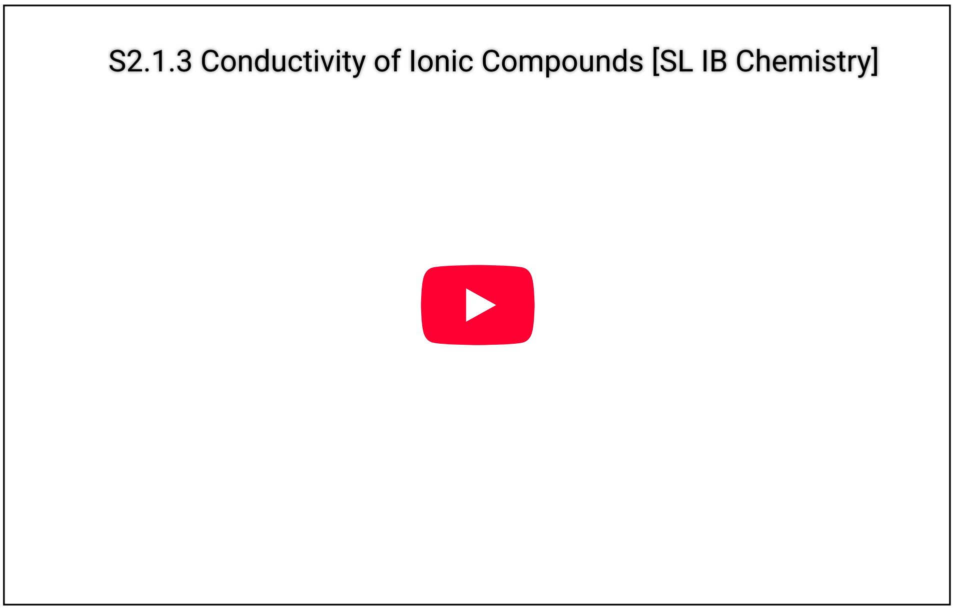


Figure 2. Electrical circuit showing when ionic compounds are electrically conductive.

More information for figure 2

Watch **Video 1** to help you visualise the conductivity of ionic compounds in the states of matter listed above.



Video 1. Visualise the conductivity of ionic compounds in the solid, molten and aqueous states.

The box below highlights common misconceptions of students when it comes to the conductivity of ionic compounds.

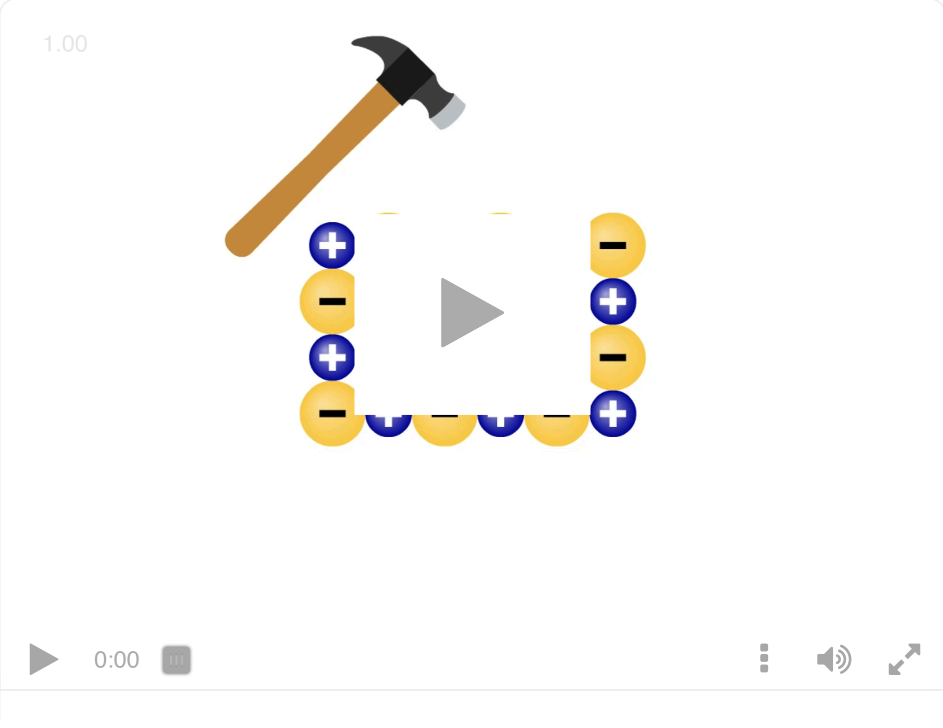
Study skills

Conductivity refers to the mobility of charged particles. It is very common for students to misinterpret conductivity to be exclusively due to electrons, however, in the case of aqueous ionic compounds, the conductivity of the solutions is from the mobility of the ions in solutions, **not** electrons.

Brittleness

Have you ever noticed if you have a large piece of table salt it can easily be ground into smaller pieces? This happens because sodium chloride is brittle. This is the final characteristic property of ionic compounds. While ionic bonds are strong, ionic substances fracture easily. These properties can be explained by taking a close look at the lattice structure as shown in **Interactive 2**.

When a stress is applied (caused by dropping or hitting with a hard object), layers within the lattice shift in response to that stress. As layers shift, ions become adjacent to like charges. These like charges experience an electrostatic repulsion and layers separate from one another, which forces the lattice to break into pieces. This is why ionic compounds, like sodium chloride in table salt, are brittle.



Interactive 2. Ionic Lattice Shifting in Response to an Applied Stress.

More information for interactive 2

The box below provides some examples of methods to collect qualitative and quantitative data for the properties of ionic compounds discussed in this section.

Practical skills

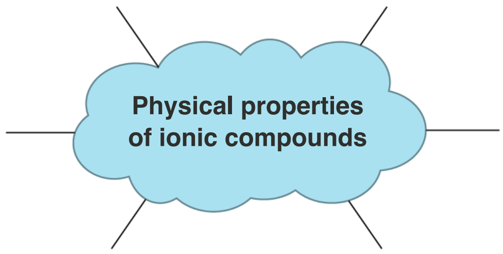
Inquiry 2: Collecting and processing data — Collecting data

There are qualitative and quantitative methods of collecting data to study the physical properties of ionic compounds. Below are examples of a few ways you could collect qualitative and quantitative data to study some of these physical properties.

	Qualitative data	Quantitative data
Meltin g point	Observe whether or not the substance melts when placed into the flame of a Bunsen burner.	Observe what temperature the substa melts at using a melting point appara
Solubi lity in water	Observe whether a small amount of the solute forms a homogeneous mixture when added to water.	Use a conductivity meter to measure h conductivity of the solution compares t water. If the ionic compound is colour spectrophotometer can also be usec measure the solubility.

	Qualitative data	Quantitative data
Electrical conductivity	Observe whether a lightbulb of a simple electrical circuit lights up.	Use a conductivity meter to measu conductivity of solution.
Brittleness	Observe (macroscopically with eyes or microscopically with microscope) the ease at which cracks and fractures form on the crystal's surface.	Use a structures and materials teste measures the force at which the mat fractures.

Drag and drop the boxes in **Interactive 3** to see whether you can recall all the physical properties of ionic compounds.



conducts electricity in solid state	low volatility	low melting points	none are soluble in water
conducts electricity in molten state	high volatility	high melting points	some are soluble in water
conducts electricity in aqueous state	brittle	flexible	all are soluble in water

Check

Interactive 3. Summary of the Physical Properties of Ionic Compounds.

More information for interactive 3

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

According to the USGS (https://www.usgs.gov/faqs/why-ocean-salty#:~:text=The%20concentration%20of%20salt%20in,be%20about%20120%20million%20tons.), there are about 120 000 000 tons of salt in a cubic mile of seawater (which is equivalent to 1.09×10^{11} kg in 4.2 km^3 of seawater) and more than 90% of this salt is sodium chloride, an ionic compound. The world's oceans are a vast source of biodiversity but they are at risk from a variety of threats such as plastic pollution and overfishing.

8 June is World Oceans Day (https://www.un.org/en/observances/oceans-day) which aims to increase the public's perception of the threats posed to the world's oceans. What can you do at your school or in your local community to raise awareness of World Oceans Day?

In the following activity you will check your understanding of the physical properties of the ionic compounds covered in this section.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning**; Thinking skills — Combining different ideas in order to create new understanding
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity for steps 1—5, pairs for step 6

1. Looking at the chemical formula for the substances below. Identify which ones you suspect are ionic. Explain how you made your selection.
(a) $MgCl_2$
(b) CO_2
(c) NH_4Br
(d) Na
2. Research the melting point of each of the substances above. Comment on how you might use the melting point to confirm or refute your selection above.
3. Explain why ionic compounds are brittle using a diagram to support your answer.

4. Sketch a diagram showing what you think happens at the atomic scale when sodium chloride dissolves in water.
5. Create a Venn diagram comparing solid ionic compounds to aqueous ionic compounds. Include structure, properties and diagrams.
6. Compare your work with a peer. Highlight three areas your work had in common. Highlight three areas where your work differed. Discuss these differences with your peer.